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10.1.1 Subspace iteration with a Hermitian matrix

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A 3x3 grid of colored squares on a dark gray background. The top row consists of three red squares. The middle row consists of a red square, an orange square, and a yellow square. The bottom row consists of a red square, an orange square, and a yellow square. A play button icon is centered over the middle row, specifically over the orange square. The play button is a dark gray rectangle with a white triangle pointing to the right.

to find one
eigenvector and one
associated
eigenvalue. In the

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Homework 10.1.1.1

1/1 point (graded)
You may want to start by executing `git pull` to update your directory `Assignments`.

Examine `Assignments/Week10/matlab/PowerMethod.m` which implements

```
[ lambda_0, v0 ] = PowerMethod( A, x, maxiters, illustrate, delay );
```

This routine implement the Power Method, starting with a vector `x` for a maximum number of iterations `maxiters` or until convergence, whichever comes first. To test it, execute the script in `Assignments/Week10/matlab/test_Subspaceliteration.m` which uses the Power Method to compute the largest eigenvalue (in magnitude) and corresponding eigenvector for an $m \times m$ Hermitian matrix A with eigenvalues $1, \dots, m$.

Be sure to click on "Figure 1" to see the graph that is created.

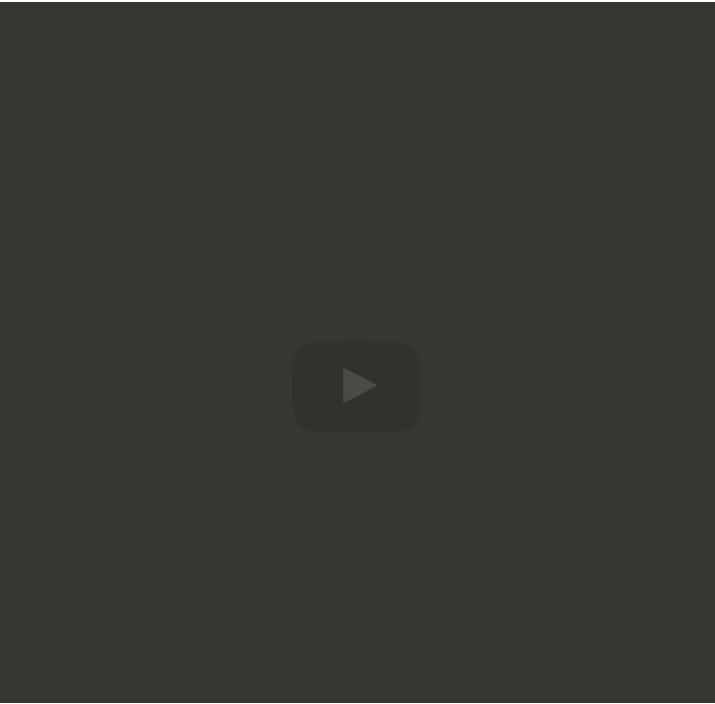
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we could say, "Hmm, we'll run the power method." We will find an eigenvector associated with the largest eigenvalue. We will call that vector that we find `x_0`.

When we pick a random vector, we start by subtracting out the component of `v` in the direction of `x_0`, which is equal to -- let's see -- `v` minus `x_0` transpose `v` times `x_0`. Well, if `x_0` of course had been normalized to have length 1. And then once we have picked

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Homework 10.1.1.2

Copy `Assignments/Week10/matlab/PowerMethod.m` into `PowerMethodLambda1.m`. Modify it by adding an input parameter `x0`, which is an eigenvector associated with λ_0 (the eigenvalue with largest magnitude).

```
[ lambda_1, v1 ] = PowerMethodLambda1( A, x, v0, maxiters, illustrate, delay )
```

The new function should subtract this vector from the initial random vector as in the above algorithm.

Modify the appropriate line in `Assignments/Week10/matlab/test_SubspaceIteration.m`, changing (0) to (1), and use it to examine the convergence of the method.

What do you observe?

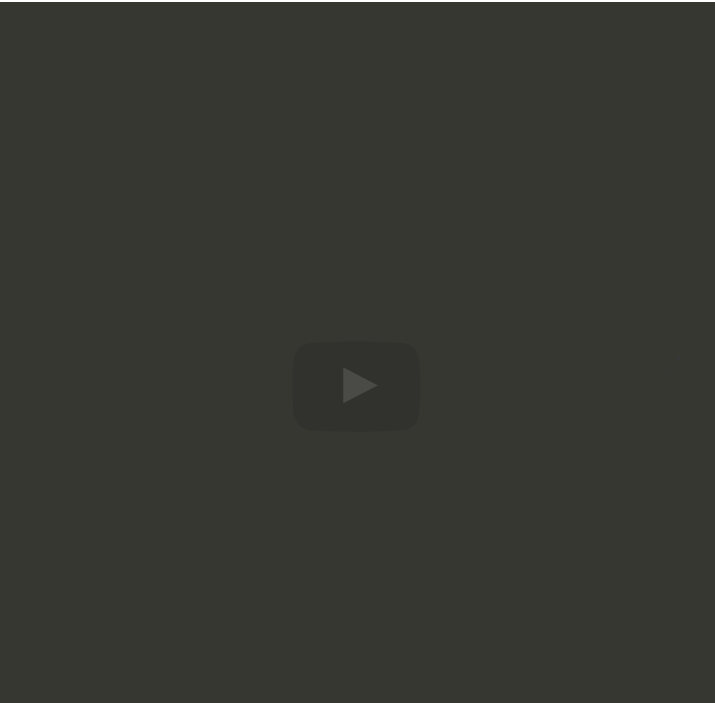
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in on when we were not subtracting out the component in the direction of `x_0`. Alright? So how can we then force that component out every time. Well, I guess I kind of gave it away. We can force that component out every time. We can say, "Hmm, before we normalize this vector let's go ahead and be diligent and subtract out again any component in the direction of `x_0` that has snuck in." Oops I should have had Hermitian

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Homework 10.1.1.3

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Copy `PowerMethodLambda1.m` into `Assignments/Week10/matlab/PowerMethodLambda1Reorth.m` and modify it to reorthogonalize with respect to `x0`:

```
[ lambda_1, v1 ] = PowerMethodLambda1Reorth( A, x, v0, maxiters,
illustrate, delay );
```

You can test your implementation by commenting out the obvious statement(s) in [Assignments/Week10/matlab/test_SubspaceIteration.m](#), changing (0) to (1), and use it to examine the convergence of the method.

What do you observe?

☒ done



Solution:

For a complete solution, see [the notes](#).

There is a video in the notes.

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Answers are displayed within the problem

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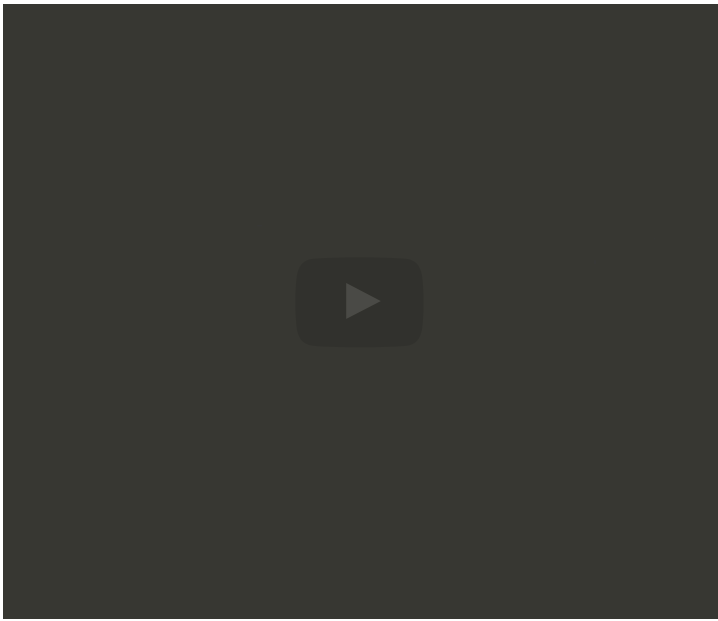
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...now here I'm
ignoring the fact that
this also returns R
because we're never
even
going to use that
matrix R. But you
know for, maybe for
completeness I
should say
we get the matrix $v_0^{(0)}$ $v_1^{(0)}$ comma R
equal to that because
typically our routine
for computing the QR
factorization of a
matrix also
returns our matrix R.
Then down here what
we could do is say,
"Hmm, let's create
temporary vectors
 v_0 v_1 by
simultaneously

...vectors to be of
length one, so we
don't need to divide.
And then like the one



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Homework 10.1.1.4

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you
would get by looking
at $v_1(k)$ Hermitian A
 $v_1(k)$. And one
elegant way of
kind of looking at that
simultaneously is to
say, "Well, if we look
at the matrix
that comes from
doing $v_0(k)$ $v_1(k)$
transpose times A
times the $v_0(k)$
 $v_1(k)$
like that, what you
expect is that that
matrix, it's obviously
a 2 by 2 matrix,



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